ISESE
FORM TWO EXAMINATION
CHEMISTRY MARKING GUIDE

FORM TWO AND FORM FOUR PRE NECTA SERIES
ISESE TIME TABLE FOR GROUP II
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ALL SUBJECTS	DATE
SERIES - 01	03rd August 2026
SERIES - 02	10th August 2026
SERIES – 03	17th August 2026
SERIES - 04	24th August 2026
SERIES – 05	31st August 2026

NB# SERIES No.06 – No.10 (SEPTEMBER – OCTOBER)

SECTION A (15 Marks)

1. *(1 mark for each correct alternative)*

I	II	III	IV	V	VI	VII	VIII	IX	X
B	C	B	C	B	B	C	B	C	C

2. *(1 mark for each correct match)*

I	II	III	IV	V
D	F	C	B	G

SECTION B (70 Marks)

3. *(2 marks for each item)*

- **(i) Why Chemistry is the study of matter:** Chemistry deals directly with the composition, structure, properties, and regular behavior of matter, as well as the changes that matter undergoes during chemical reactions.
- **(ii) Two professional career fields:** *(Any two)*: Medicine/Pharmacology, Chemical Engineering, Agriculture, Forensic Science, Metallurgy, or Biochemistry.
- **(iii) Benefit of chemical fertilizers to agriculture:** They replenish essential nutrients (like Nitrogen, Phosphorus, and Potassium) in the soil, boosting crop growth and increasing food production efficiency.
- **(iv) One negative effect of bad waste disposal:** It causes soil degradation, water pollution (leading to the death of aquatic life), and toxic chemical accumulation in the human food chain.
- **(v) Chemical product vs. Raw material:** A *chemical raw material* is an unprocessed starting substance used to begin an industrial process (e.g.,

crude oil, rock salt). A *chemical product* is the final substance manufactured after chemical processing (e.g., plastic, soap).

4. *(2 marks for each item)*

- **(i) Specialized safety chamber: Fume cupboard** (or Fume hood).
- **(ii) Function of goggles and gloves:** *Safety goggles* protect the eyes from dangerous chemical splashes or flying debris. *Laboratory gloves* protect the hands from chemical burns, corrosive substances, and heat.
- **(iii) Why smelling or tasting directly is forbidden:** Many laboratory chemicals are highly poisonous, toxic, or corrosive, and can cause respiratory damage, organ failure, or immediate tissue burns.
- **(iv) Two apparatus for measuring fixed liquid volumes accurately: Pipette** and **Volumetric flask**. *(Note: Measuring cylinders can be accepted for general volume measurement)*.
- **(v) Meaning of skull and crossbones sign:** It indicates that the chemical is **Toxic / Poisonous** and can cause death or severe illness if swallowed, inhaled, or absorbed through the skin.

5. *(2 marks for each item)*

- **(i) Type of flame with excess oxygen: Non-luminous flame**.
- **(ii) Hottest zone of the non-luminous flame:** The **outermost zone** (or pale-blue zone/region of complete combustion).
- **(iii) Why the soot-producing flame is undesirable:** The soot (unburnt carbon particles) deposits on glass apparatus, turning them black, obscuring visibility, and causing uneven heating.
- **(iv) Two elements required for a fuel to burn with a flame: Carbon** and **Hydrogen** (forming hydrocarbons in the fuel that burn in the presence of oxygen).
- **(v) Main difference in structural zones:** A *luminous flame* has four distinct zones (outermost non-luminous zone, luminous zone, central dark zone, blue zone at the base), while a *non-luminous flame* has three distinct zones (outermost pale-blue zone, middle greenish-blue zone, innermost dark zone).

6. *(2 marks for each item)*

- **(i) Identification of materials:** The physically joined material is a **Mixture**. The pure substance that cannot be broken down further is an **Element**.
- **(ii) Two differences between a compound and a mixture:** 1. In a *compound*, components are chemically combined, whereas in a *mixture*, they are physically combined.
  2. A *compound* has fixed proportions of elements, whereas a *mixture* has variable proportions of components.

- **(iii) Physical property for fractional distillation:** Differences in **boiling points** of the miscible liquids.
- **(iv) Why sublimation separates iodine from sand:** Iodine changes directly from a solid to a gas phase upon heating (sublimes), leaving the non-sublimable sand behind in the container.
- **(v) Domestic example of a homogeneous mixture:** Salt solution, sugar solution, clean drinking water, or hot tea.

7. *(2 marks for each item)*

- **(i) Three subatomic particles: Protons**, **Neutrons**, and **Electrons**.
- **(ii) Location and charge of an electron:** It is located **outside the nucleus** (moving around energy levels/shells) and carries a **Negative charge**
- **(iii) Definition of Isotopes:** Atoms of the same element with the same atomic number (number of protons) but different mass numbers due to a different number of neutrons.
- **(iv) Mass number configuration: 18** (Mass Number = Protons + Neutrons = 8 + 10 = 18).
- **(v) Why elements in the same Group have similar properties:** Because they possess the exact same number of valence electrons (electrons in their outermost energy shell), which dictates how they bond and react chemically.

8. *(2 marks for each item)*

- **(i) Bonding type in Sodium Chloride ($NaCl$): Ionic bond** (or Electrovalent bond).
- **(ii) Bonding type in Carbon Tetrachloride (CCl_4): Covalent bond**.
- **(iii) Why ionic compounds have high melting points:** They possess strong electrostatic forces of attraction holding the oppositely charged ions together in a giant crystal lattice, requiring a massive amount of heat energy to break down.
- **(iv) Electricity conduction states:** In an *aqueous state* (or molten state), the crystal lattice breaks down, leaving ions free to move and conduct an electric current. In a *solid state*, the ions are locked in fixed positions and cannot move freely.
- **(v) Electronic condition for stability:** An atom must attain a stable configuration of 2 electrons in its valence shell (**Duplet rule**) or 8 electrons in its valence shell (**Octet rule**), matching the electron configuration of a noble gas.

9. *(2 marks for each item)*

- **(i) Type of chemical reaction: Neutralization reaction**.
- **(ii) Definition of an Acid:** A substance that dissociates/ionizes in water to yield hydrogen ions as the only positively charged ions.

- **(iii) Two general properties of basic substances:** *(Any two)*: They have a bitter taste; They feel soapy or slippery when touched; They turn red litmus paper blue; They react with acids to form salt and water only.
- **(iv) Normal salt vs. Acidic salt:** A *normal salt* is formed when all the replaceable hydrogen ions of an acid are completely replaced by a metal or ammonium ion. An *acidic salt* is formed when the replaceable hydrogen ions are only partially replaced.
- **(v) Commercial industrial use of salts:** *(Any one)*: In food preservation (curing meat); Manufacturing of soap; Dyeing fabrics in textile industries; Softening hard water (using sodium carbonate).

SECTION C (15 Marks)

10. *(Each major requirement carries 5 marks)*

- **Definition and Three Indicators of a Chemical Reaction (5 Marks):**
  - *Definition:* A chemical reaction is a process whereby one or more substances (reactants) undergo a chemical transformation to form entirely new substances (products) with different properties.
  - *Three Indicators:* Evolution of a gas (effervescence), change in color, formation of a precipitate, or a distinct temperature change (heat absorption or evolution).

- **Combination vs. Decomposition Reactions (5 Marks):**
  - *Combination Reaction:* A reaction where two or more simple reactants chemically fuse to form a single complex product. *Example:* Burning magnesium ribbon in oxygen to form magnesium oxide or manufacturing ammonia.
  - *Decomposition Reaction:* A reaction where a single complex compound is broken down into two or more simpler substances, often by heat or electricity. *Example:* Thermal decomposition of limestone (calcium carbonate) into lime and carbon dioxide ($CaCO_3 \rightarrow CaO + CO_2$).
- **Two Factors that Accelerate Chemical Reactions (5 Marks):**
  1. **Temperature:** Increasing the temperature gives reactant particles more kinetic energy, causing them to move faster and collide more frequently and effectively, thus increasing the reaction rate.
  2. **Concentration of Reactants:** Increasing the concentration means there are more reactant particles crowded into the same volume, increasing the frequency of collisions between particles.
  3. *(Alternative valid factors)*: Presence of a catalyst, surface area of solid reactants, or exposure to light (for photochemical reactions).